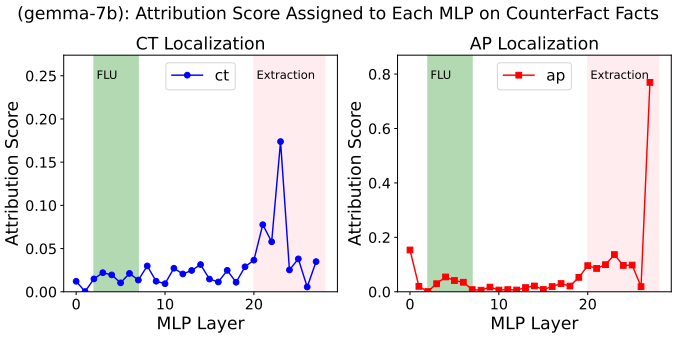

Figure 14: Attribution scores on MLPs on CounterFact facts for Gemma-7B.

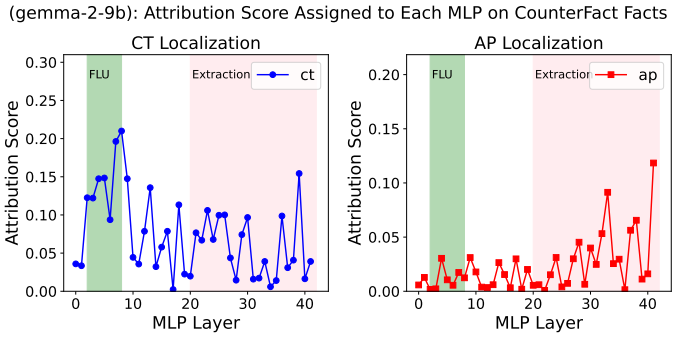

Figure 15: Attribution scores on MLPs on CounterFact facts for Gemma-2-9B.

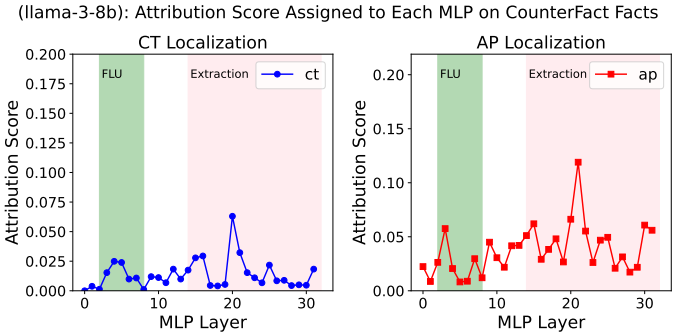

Figure 16: Attribution scores on MLPs on CounterFact facts for Llama-3-8B.

A.4 Weight Masking

In this section we employ weight masking to quantify the size of weight updates needed to unlearn/edit facts, for more direct comparisons. Our loss function $L = \lambda_1 L_{\text{forget}} + \lambda_2 L_{\text{retain}} + \lambda_3 L_{\text{SFT}} + \lambda_4 L_{\text{reg}}$ now includes an L1 regularization term to control the sparsity. We empirically evaluate how a learned binary mask over individual weights of the localized components can produce editing/unlearning, and vary the size of this mask. "Manual Interp" refers to the FLU localization technique for all the following results in this section.

A.4.1 Unlearning Sports

We show standard evaluations across a sweep of discretization thresholds, which directly corresponds to the size of the model edit. Figure 17 shows the accuracy on the forget and retain sets for unlearning basketball across different edit sizes. Here, we see all methods being effective in

unlearning basketball facts while retaining all other facts. While AP and CT localizations cause the model to have zero accuracy on the in-distribution set with much fewer masked weights needed, when checking for generalization using a multiple-choice format we clearly see that only manual localization has successfully generalized the unlearning of basketball facts (Figure 17, right).

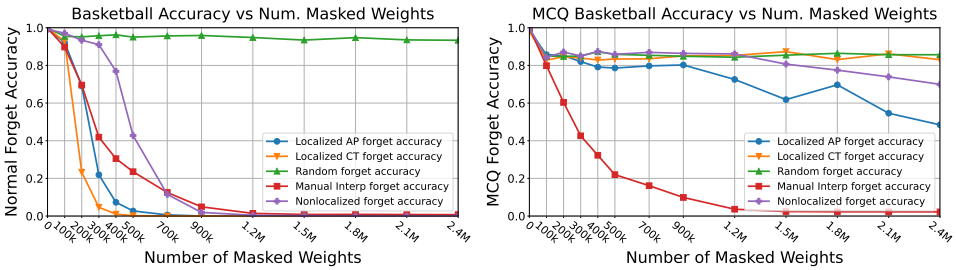

Figure 17: **(Left)** Testing the models' unlearning of basketball athletes against the number of weights masked. **(Right)** Testing the models' unlearning of basketball athletes against the number of weights masked, in the MCQ prompt format.

We find similar results when testing for performance degradation on MMLU (because we have to evaluate many model variations, we use a smaller MMLU test set from Polo et al. (2024)). While all localized methods perform well when evaluated normally (Figure 18, left), Figure 18 (right) shows manual localization generalizes for minimizing loss of MMLU capabilities while unlearning sports facts in the MCQ format compared to the other methods.

A.4.2 Editing Athletes

For editing the subset of athletes, Figure 19 shows that causal tracing localization causes the model to have 0% accuracy on the forget set, and FLU and nonlocalized editing cause the model to have near guessing rate (33%) accuracy. However, only manual localization minimizes loss of capabilities while editing the athlete subset (Figure 20).

Furthermore, no other method completely generalizes this unlearning to the MCQ prompt format (Figure 19), and manual localization remains superior in minimizing loss of capabilities while unlearning the athlete subset (Figure 20, right).

A.4.3 Editing CounterFact

We find similar results for editing on the CounterFact dataset. However, we find minimal difference in MMLU accuracy in all methods at all numbers of masked weights. Thus, we instead report the maintain and forget accuracies of these facts at different discretization thresholds in Figure 21.

Additionally, we report a comparison of all localizations across discretization thresholds for normal and MCQ forget sets in Figure 22 and Figure 23. We see that FLU outperforms all other methods of localization in preserving maintain accuracy while decreasing forget accuracy.

We perform additional adversarial analysis of accuracies across different discretization thresholds. We report the "paraphrase" and "neighborhood" adversarial results in Figure 24 and Figure 25, but find no significant results.

A.5 Mechanism Weight Analysis

We analyze the actual components localized by each localization type and our baselines, for the CounterFact editing task. We seek to demonstrate that the OT localizations and baselines target extraction mechanisms rather than just the FLU mechanisms.

First, in Table 2, we compare the parameter counts of the part of each mechanism that is present in each localization. Table 2 shows that causal tracing and attribution patching both have the potential to modify a considerable proportion of the extraction heads and extraction MLPs.

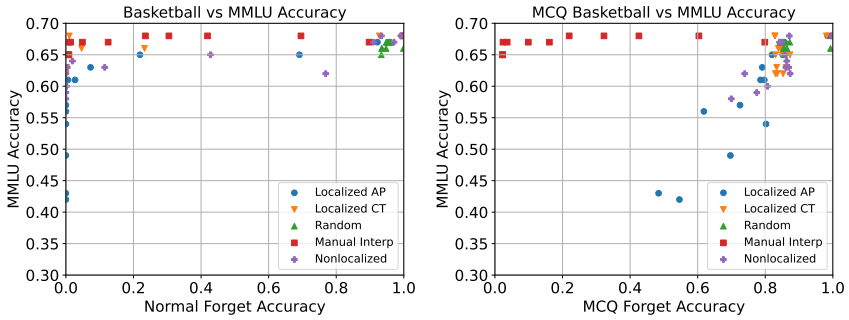

Figure 18: Unlearning basketball facts. **(Left)** Measuring MMLU and forget set performance across different discretization thresholds. **(Right)** Measuring MMLU and MCQ forget set performance across different discretization thresholds.

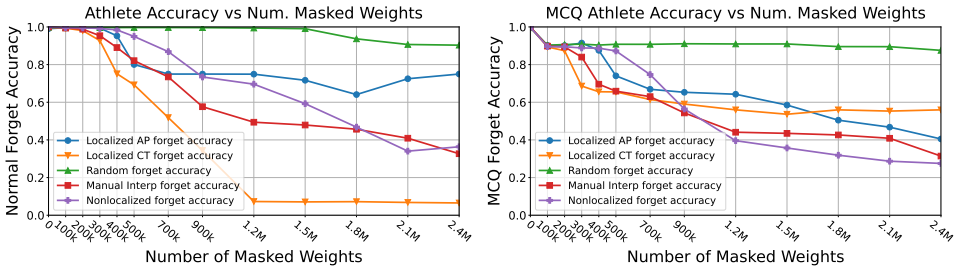

Figure 19: Editing subset of athletes. **(Left)** Measuring accuracy on the forget set. **(Right)** Measuring accuracy on the forget set in the MCQ prompt format.

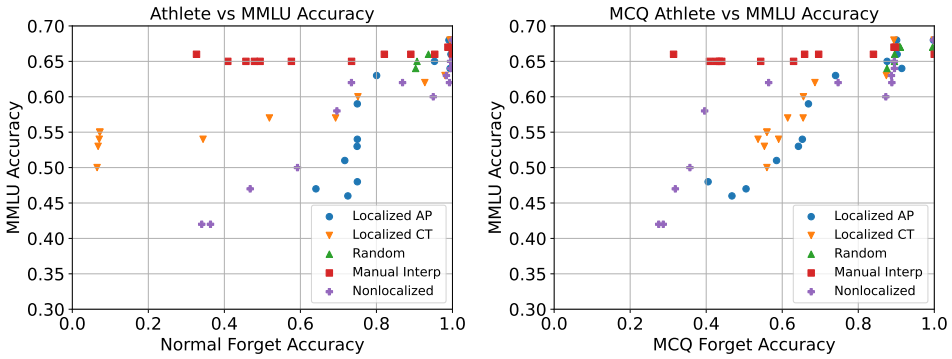

Figure 20: Editing subset of athletes. **(Left)** Measuring MMLU and forget set performance across different discretization thresholds. **(Right)** Measuring MMLU and MCQ forget set performance across different discretization thresholds.

Then, in Table 3, we compare the proportion of each mechanism that is masked when using a localized weight mask and discretizing to about 6 million weights. This is one approximate metric for how much each mechanism is modified by the localized editing. Table 3 demonstrates that attribution patching, causal tracing, and nonlocalized editing all modify a higher proportion of the extraction head/MLP weights than the fact lookup mechanism weights.

This supports our argument that OT methods target high logit-diff extraction mechanisms, rather than the fact lookup mechanisms that enrich the latent stream with the correct attributes, which decreases the robustness of edits/unlearning. It is important to note that since our FLU localization is based on our discovered mechanisms, this does not serve as an evaluation of FLU (since by definition FLU localization will only localize the FLU mechanism), but rather only of causal tracing and attribution patching.